Data Mining Trends and Research.txt

```
📚 DATA MINING: FUTURE, APPLICATIONS & SOCIETY
│
├────────────────────────────────────────────
│ 1. MINING COMPLEX DATA TYPES
├────────────────────────────────────────────
│
├── A. Sequence Data
│   │
│   ├── Time-Series Data
│   │   • Ordered numeric values recorded at equal intervals.
│   │   • Examples: Stock prices, weather records, ECG signals.
│   │   • Similarity Search:
│   │       - Find subsequences similar to a query sequence.
│   │       - Uses DFT and DWT for dimensionality reduction.
│   │   • Trend Analysis Components:
│   │       1. Long-Term Trend
│   │       2. Cyclic Movement
│   │       3. Seasonal Variation
│   │       4. Random Movement
│   │
│   ├── Symbolic Sequences
│   │   • Sequence of events/nominal values.
│   │   • Equal intervals are not required.
│   │   • Gaps usually not important.
│   │   • Examples:
│   │       - Website clickstreams
│   │       - Shopping history
│   │       - Event logs
│   │
│   └── Biological Sequences
│       • DNA and Protein sequences.
│       • Extremely long and semantically rich.
│       • Gaps are important.
│       • Tools: BLAST
│       • Models: Hidden Markov Models (HMM)
│
├── B. Graphs & Networks
│   │
│   ├── Graph Structure
│   │   • Nodes (Objects)
│   │   • Edges (Relationships)
│   │
│   ├── Homogeneous Network
│   │   • Same type of nodes and links.
│   │   • Example: Facebook friendship network.
│   │
│   ├── Heterogeneous Network
│   │   • Different types of nodes and links.
│   │   • Example:
│   │       Doctor ↔ Patient ↔ Disease ↔ Medicine
│   │
│   └── Important Network Properties
│       • Power Law Distribution:
│           Few nodes have many connections.
│       • Small World Phenomenon:
│           Few hops connect most people.
│       • Densification:
│           Networks become denser over time.
│       • Shrinking Diameter:
│           Average distance decreases.
│
├── C. Other Data Modalities
│   │
│   ├── Spatial Data
│   │   • Geographic information.
│   │   • Stored as Vector and Raster data.
│   │   • Uses Spatial OLAP and Data Cubes.
│   │
│   ├── Spatiotemporal Data
│   │   • Space + Time together.
│   │   • Examples:
│   │       GPS tracking
│   │       Hurricane movement
│   │       Vehicle routes
│   │
│   ├── Cyber-Physical Systems (CPS)
│   │   • Physical devices connected with software.
│   │   • Examples:
│   │       Smart traffic systems
│   │       Smart factories
│   │   • Require real-time response.
│   │
│   ├── Multimedia Data
│   │   • Images, Videos, Audio.
│   │   • Uses content-based retrieval.
│   │   • Similarity search is important.
│   │
│   ├── Text Mining
│   │   • Extract knowledge from text.
│   │   • Applications:
│   │       Sentiment Analysis
│   │       Entity Extraction
│   │       Topic Detection
│   │
│   ├── Web Mining
│   │   • Web Content Mining:
│   │       Text, Images, Videos.
│   │
│   │   • Web Structure Mining:
│   │       Hyperlinks and HTML structure.
│   │
│   │   • Web Usage Mining:
│   │       User clickstream analysis.
│   │
│   └── Data Streams
│       • Continuous and fast-flowing data.
│       • Infinite in nature.
│       • Single scan algorithms required.
│       • Uses Sliding Window processing.
│
├────────────────────────────────────────────
│ 2. ADVANCED METHODOLOGIES
├────────────────────────────────────────────
│
├── Regression
│   • Predicts continuous numeric values.
│   • Example:
│       House price prediction.
│
├── Generalized Linear Models (GLM)
│   • Extension of regression.
│   • Handles categorical outcomes.
│   • Example:
│       Logistic Regression.
│
├── ANOVA
│   • Analysis of Variance.
│   • Compares means of multiple groups.
│   • Determines if differences are significant.
│
├── Factor Analysis
│   • Finds hidden (latent) factors.
│   • Reduces many variables into key factors.
│   • Example:
│       Intelligence from test scores.
│
└── Survival Analysis
    • Predicts probability of survival until time t.
    • Applications:
        Medical survival prediction
        Machine failure prediction
│
├────────────────────────────────────────────
│ 3. THEORETICAL FOUNDATIONS
├────────────────────────────────────────────
│
├── Data Reduction Theory
│   • Data mining reduces data size.
│   • Produces compact representations.
│
├── Data Compression Theory
│   • Mining compresses data into:
│       Rules
│       Trees
│       Clusters
│   • Based on MDL Principle
│     (Minimum Description Length).
│
├── Microeconomic View
│   • Pattern is useful only if it creates value.
│   • Focus on utility maximization.
│
└── Pattern Discovery /
    Inductive Databases
    • Database stores:
        Raw Data
        Discovered Patterns
│
├────────────────────────────────────────────
│ 4. APPLICATION DOMAINS
├────────────────────────────────────────────
│
├── Financial Data Analysis
│   • Loan payment prediction.
│   • Credit risk assessment.
│   • Customer profiling.
│   • Money laundering detection.
│   • Uses prediction cubes.
│
├── Retail & Telecommunications
│   • Targeted marketing.
│   • Customer segmentation.
│   • Loyalty analysis.
│   • Product recommendation.
│   • Customer retention.
│   • Fraud detection.
│
├── Science & Engineering
│   • Climate modeling.
│   • Astronomy.
│   • Gene analysis.
│   • Bioinformatics.
│
└── Intrusion Detection Systems (IDS)
    │
    ├── Signature-Based Detection
    │   • Uses known attack signatures.
    │   • Accurate for known attacks.
    │   • Cannot detect zero-day attacks.
    │
    └── Anomaly-Based Detection
        • Learns normal behavior.
        • Flags unusual activities.
        • Detects unknown attacks.
        • High False Positive rate.
│
├────────────────────────────────────────────
│ 5. RECOMMENDER SYSTEMS
├────────────────────────────────────────────
│
├── Goal
│   • Predict user preference.
│   • Recommend useful items.
│
├── Content-Based Filtering
│   • Recommends items similar to
│     previously liked items.
│   • Uses:
│       Genre
│       Keywords
│       Features
│       Attributes
│
├── Collaborative Filtering (CF)
│   │
│   ├── Memory-Based CF
│   │   • Uses complete rating data.
│   │   • k-Nearest Neighbour (kNN)
│   │   • Similarity Measures:
│   │       Pearson Correlation
│   │       Cosine Similarity
│   │
│   └── Model-Based CF
│       • Builds predictive models.
│       • Uses:
│           Bayesian Networks
│           Clustering
│           Machine Learning
│
├── Evaluation Errors
│   │
│   ├── False Negative
│   │   • Good item not recommended.
│   │
│   └── False Positive
│       • Bad item recommended.
│       • Most undesirable error.
│
└── Hybrid Systems
    • Combine multiple recommenders.
    • Improve accuracy.
    • Famous Example:
        Netflix Prize Solution.
│
├────────────────────────────────────────────
│ 6. PRIVACY-PRESERVING DATA MINING
│    (PPDM)
├────────────────────────────────────────────
│
├── Purpose
│   • Discover patterns while protecting
│     personal information.
│
├── Randomization
│   • Add random noise to data.
│   • Hides individual values.
│   • Preserves aggregate statistics.
│
├── k-Anonymity
│   • Each record looks identical to
│     at least (k-1) other records.
│
│   • Techniques:
│       Generalization
│       Suppression
│
│   • Example:
│       Age 23 → Age 20-30
│
├── l-Diversity
│   • Extension of k-Anonymity.
│   • Each group must contain at least
│     l diverse sensitive values.
│   • Prevents inference attacks.
│
├── Distributed Privacy Preservation
│   • Data distributed across sites.
│
│   • Horizontal Partition:
│       Different rows.
│
│   • Vertical Partition:
│       Different attributes.
│
│   • Uses cryptographic protocols.
│
└── Differential Privacy
    • Strongest privacy model.
    • Output remains nearly same
      whether a person's data exists
      or not.
    • Provides mathematical guarantee
      of privacy.
│
├────────────────────────────────────────────
│ 7. DATA MINING & SOCIETY
├────────────────────────────────────────────
│
├── Ubiquitous Data Mining
│   • Data mining is everywhere.
│   • Search engines.
│   • Recommendation systems.
│   • Social media.
│   • Online shopping.
│
├── Invisible Data Mining
│   • Works in background.
│   • Users often don't notice it.
│   • Examples:
│       Spam filters
│       Search suggestions
│       Personalized ads
│
└── Major Concerns
    • Privacy
    • Security
    • Ethical use of data
│
├────────────────────────────────────────────
│ 8. FUTURE TRENDS
├────────────────────────────────────────────
│
├── Scalable Data Mining
│   • Handle huge datasets.
│
├── Interactive Mining
│   • User-guided mining process.
│
├── Constraint-Based Mining
│   • Discover only useful patterns.
│
├── Cloud Mining
│   • Distributed processing.
│
├── Search Engine Integration
│   • Better search and ranking.
│
├── Real-Time Mining
│   • Instant decision making.
│
└── Stream Mining
    • Continuous processing of
      high-speed data streams.
```

Web & Social Network Mining.txt

```
🌐 WEB MINING & SOCIAL NETWORK MINING
│
├─────────────────────────────────────
│ 1. WEB MINING FUNDAMENTALS
├─────────────────────────────────────
│
├── Definition
│   • Application of data mining techniques
│     on web data.
│   • Purpose:
│       Discover useful knowledge,
│       patterns and relationships.
│
├── Data Mining vs Web Mining
│   │
│   ├── Data Mining
│   │   • Structured data.
│   │   • Relational databases.
│   │   • Rows, Columns, Keys.
│   │
│   │   Example:
│   │     Student database
│   │     Bank database
│   │
│   └── Web Mining
│       • Semi-structured data.
│       • Unstructured data.
│       • HTML pages.
│       • Images, videos, text.
│       • Hyperlinks.
│
│       Example:
│         Google Search
│         News Websites
│
├── Web Characteristics
│   │
│   ├── Huge & Wide Coverage
│   │   • Massive amount of information.
│   │   • Almost every topic available.
│   │
│   ├── Semi-Structured
│   │   • HTML-based documents.
│   │
│   ├── Highly Linked
│   │   • Connected through hyperlinks.
│   │
│   ├── Noisy
│   │   • Ads
│   │   • Menus
│   │   • Copyright notices
│   │
│   ├── Redundant
│   │   • Same information repeated
│   │     on multiple websites.
│   │
│   └── Dynamic
│       • New pages continuously created.
│       • Existing pages updated frequently.
│
├─────────────────────────────────────
│ 2. THREE CATEGORIES OF WEB MINING
├─────────────────────────────────────
│
│                WEB MINING
│                    │
│       ┌────────────┼────────────┐
│       ▼            ▼            ▼
│
│  Content Mining  Usage Mining  Structure Mining
│
├─────────────────────────────────────
│ A. WEB CONTENT MINING
├─────────────────────────────────────
│
├── Definition
│   • Extract information from the
│     actual content of web pages.
│
├── Data Sources
│   • Text
│   • Images
│   • Videos
│   • Audio
│   • Graphs
│
├── Goal
│   • Determine relevance of content
│     to search queries.
│
├── Techniques
│   │
│   ├── Classification
│   │   • Place documents into
│   │     predefined categories.
│   │
│   │   Example:
│   │     Science
│   │     Politics
│   │     Sports
│   │
│   ├── Clustering
│   │   • Group similar documents.
│   │   • No predefined labels needed.
│   │
│   ├── Summarization
│   │   • Reduce document length.
│   │   • Keep main ideas intact.
│   │
│   └── Visualization
│       • Graphical representation
│         of information.
│
│       Examples:
│         Word Clouds
│         Scatter Plots
│         Heat Maps
│
├── Benefits
│   • Better search results.
│   • Easier information retrieval.
│   • Topic discovery.
│
└── Applications
    • Search Engines
    • Digital Libraries
    • News Portals

─────────────────────────────────────

📊 B. WEB USAGE MINING
│
├── Definition
│   • Discover user browsing patterns.
│   • Understand user behavior.
│
├── Main Data Source
│   • Server Log Files (Web Logs)
│
├── Information Stored
│   • Clicks
│   • Page Views
│   • Form Submissions
│   • User Sessions
│
├── Goals
│   • Predict future behavior.
│   • Improve websites.
│   • Support personalization.
│
├── Three Core Phases
│   │
│   ├── 1. Data Preprocessing
│   │   • Data Cleaning
│   │   • Data Integration
│   │   • Data Transformation
│   │
│   │   Result:
│   │   Raw logs → Structured logs
│   │
│   ├── 2. Pattern Discovery
│   │   • Apply mining algorithms.
│   │   • Find hidden patterns.
│   │
│   │   Techniques:
│   │     Classification
│   │     Clustering
│   │     Association Rules
│   │
│   └── 3. Pattern Analysis
│       • Evaluate patterns.
│       • Predict future behavior.
│
├── Example
│   • User visits Laptop page.
│   • Then visits Mouse page.
│   • Pattern discovered:
│       Laptop → Mouse
│
└── Applications
    • Customer Analysis
    • Personalized Websites
    • Marketing
    • Product Recommendation

─────────────────────────────────────

🔗 C. WEB STRUCTURE MINING
│
├── Definition
│   • Analyze hyperlink structure
│     of the web.
│
├── Web Graph Model
│
│   • Nodes = Web Pages
│   • Edges = Hyperlinks
│
├── Objectives
│   • Improve search quality.
│   • Reduce irrelevant results.
│   • Better indexing.
│
├── Important Terms
│   │
│   ├── In-Degree
│   │   • Number of incoming links.
│   │
│   │   Formula:
│   │     Links pointing TO page P
│   │
│   ├── Out-Degree
│   │   • Number of outgoing links.
│   │
│   │   Formula:
│   │     Links leaving page P
│   │
│   ├── Directed Path
│   │   • Sequence of directed links.
│   │
│   ├── Shortest Path
│   │   • Minimum links required
│   │     to reach destination.
│   │
│   └── Diameter
│       • Longest shortest path
│         in web graph.
│
├── Hubs & Authorities
│   │
│   ├── Good Hub
│   │   • Points to many good authorities.
│   │
│   │   Example:
│   │     Resource directory page.
│   │
│   └── Good Authority
│       • Receives links from
│         many good hubs.
│
│       Example:
│         Official Wikipedia page
│         Research institution page
│
└── Importance
    • Search Engine Ranking
    • Link Analysis
    • Website Importance Measurement

─────────────────────────────────────

📱 3. SOCIAL NETWORK MINING
─────────────────────────────────────
│
├── Definition
│   • Extract patterns and trends
│     from social media data.
│
├── Social Network Analysis (SNA)
│
│   • Mathematical and visual mapping
│     of relationships.
│
│   Purpose:
│     Understand interactions between
│     people, groups and organizations.
│
├── Components
│   │
│   ├── Nodes
│   │   • People
│   │   • Groups
│   │   • Organizations
│   │
│   └── Links
│       • Friendships
│       • Communication
│       • Relationships
│       • Information Flow
│
├── Goals of SNA
│   • Find influential users.
│   • Detect communities.
│   • Understand information spread.
│   • Analyze social behavior.
│
├── Core Techniques
│   │
│   ├── Graph Mining
│   │
│   │   • Works on network structure.
│   │   • Extracts patterns from graphs.
│   │
│   │   Applications:
│   │     Community Detection
│   │     Influencer Detection
│   │     Fraud Detection
│   │     Anomaly Detection
│   │
│   └── Text Mining
│
│       • Works on textual content.
│
│       Data Sources:
│         Blogs
│         Reviews
│         Tweets
│         Comments
│         Messages
│
│       Applications:
│         Sentiment Analysis
│         Spam Detection
│         Trend Analysis
│         Document Classification
│
└── Examples
    • Twitter/X trend analysis
    • Facebook friendship analysis
    • YouTube comment analysis

─────────────────────────────────────

💻 4. E-COMMERCE APPLICATIONS
─────────────────────────────────────
│
├── Customer Analysis
│
│   Goal:
│     Understand customer behavior.
│
│   Questions Answered:
│     • Will customer buy?
│     • What product will be bought?
│     • Is user browsing or shopping?
│
├── Web Personalization
│
│   Definition:
│     Automatically customize website
│     according to user behavior.
│
│   Uses:
│     Page Views
│     Click History
│     Search History
│     Form Entries
│
│   Examples:
│     Amazon recommendations
│     Netflix homepage personalization
│
├── Product Search
│
│   • Analyze search logs.
│   • Discover relationships between
│     products and queries.
│
└── Product Recommendation
    • Recommend related products.
    • Based on previous searches.
    • Based on previous purchases.

    Example:
      Search Laptop
      → Recommend Mouse, Bag, Keyboard

─────────────────────────────────────

⭐ VERY IMPORTANT EXAM COMPARISONS

1. Data Mining vs Web Mining
   • Structured vs Semi/Unstructured

2. Web Content vs Usage vs Structure Mining
   • Content = Page Data
   • Usage = User Behavior
   • Structure = Hyperlinks

3. Hub vs Authority
   • Hub → Points to many authorities
   • Authority → Pointed by many hubs

4. In-Degree vs Out-Degree
   • In = Incoming Links
   • Out = Outgoing Links

5. Graph Mining vs Text Mining
   • Graph = Relationships
   • Text = Comments, Reviews, Messages

─────────────────────────────────────

🎯 MOST FREQUENTLY ASKED KEYWORDS

• Web Mining
• Web Content Mining
• Web Usage Mining
• Web Structure Mining
• Server Logs (Web Logs)
• Classification
• Clustering
• Summarization
• Visualization
• Pattern Discovery
• Pattern Analysis
• Web Graph
• In-Degree
• Out-Degree
• Diameter
• Hub
• Authority
• Social Network Analysis (SNA)
• Graph Mining
• Text Mining
• Customer Analysis
• Web Personalization
• Product Recommendation
```

Web Mining Techniques.txt

```
🌐 WEB MINING TECHNIQUES
│
──────────────────────────────
1. WHAT IS WEB MINING?
──────────────────────────────
│
├── Definition
│   • Web Mining = Applying Data Mining
│     techniques on Web data.
│
│   • Purpose:
│     Extract useful information,
│     hidden patterns and knowledge
│     from the World Wide Web.
│
├── Mining Analogy
│   • Gold Mining → Extract gold from earth.
│   • Web Mining → Extract useful knowledge
│     from web pages and web data.
│
├── Sources Used in Web Mining
│   • Web Documents
│   • Hyperlinks
│   • Web Services
│   • Server Logs
│
└── Goal
    • Discover patterns.
    • Understand users.
    • Improve search results.
    • Support business decisions.

──────────────────────────────

📊 2. TRADITIONAL DATA MINING VS WEB MINING
   (VERY IMPORTANT EXAM QUESTION)
──────────────────────────────

├── Traditional Data Mining
│
│   • Structured Data
│   • Relational Databases
│   • Tables
│   • Rows
│   • Columns
│   • Keys
│
│   Characteristics:
│   • Clean
│   • Organized
│   • Mostly Static
│
│   Example:
│     Student Database
│     Bank Database
│
└── Web Mining
    │
    • Semi-Structured Data
    • Unstructured Data
    • HTML Documents
    • Text
    • Images
    • Videos
    • Hyperlinks

    Characteristics:
    • Dynamic
    • Noisy
    • Redundant
    • Linked

    Example:
      Web Pages
      Blogs
      Social Media

──────────────────────────────

⭐ QUICK COMPARISON TABLE

Feature          Traditional DM       Web Mining
─────────────────────────────────────────────────
Data Type        Structured           Semi/Unstructured

Storage          Tables               HTML Pages

Organization     Well Organized       Messy

Nature           Static               Dynamic

Quality          Clean                Noisy

Duplication      Low                  High (Redundant)

Connectivity     Independent          Hyperlinked

Examples         Databases            Websites

──────────────────────────────

🌐 3. THREE MAIN CLASSES OF WEB MINING
──────────────────────────────

                 WEB MINING
                     │
         ┌───────────┼───────────┐
         ▼           ▼           ▼

  Structure Mining  Content Mining  Usage Mining
──────────────────────────────

🔗 A. WEB STRUCTURE MINING

├── Definition
│   • Analyze structure of hyperlinks.
│
├── Web Representation
│   • Nodes = Web Pages
│   • Edges = Hyperlinks
│
├── Purpose
│   • Understand page relationships.
│   • Improve search engines.
│
└── Example
    • Google analyzing links
      between websites.

──────────────────────────────

📄 B. WEB CONTENT MINING

├── Definition
│   • Extract information from
│     page contents.
│
├── Content Includes
│   • Text
│   • Images
│   • Videos
│   • Audio
│
├── Main Techniques
│   │
│   ├── Document Classification
│   │
│   │   Definition:
│   │   Assign documents to
│   │   predefined categories.
│   │
│   │   Example:
│   │     Science
│   │     Politics
│   │     Sports
│   │
│   ├── Document Clustering
│   │
│   │   Definition:
│   │   Group similar documents
│   │   automatically.
│   │
│   │   No predefined labels needed.
│   │
│   └── Association
│
│       Definition:
│       Discover relationships
│       among content items.
│
│       Example:
│         Users reading AI article
│         also read ML article.
│
└── Goal
    • Improve relevance of search results.

──────────────────────────────

📊 C. WEB USAGE MINING

├── Definition
│   • Analyze user browsing behavior.
│
├── Purpose
│   • Understand users.
│   • Discover browsing patterns.
│   • Predict future actions.
│
├── Data Source
│   • Server Logs
│   • Browser Logs
│   • Clickstreams
│
└── Example
    • Which pages users visit most?
    • Which products are viewed together?

──────────────────────────────

🔥 4. WEB USAGE MINING LOGS
   (MOST IMPORTANT TOPIC)
──────────────────────────────

Web Usage Mining depends on LOG FILES.

A Log = Record of user activity stored by server.

There are FOUR important logs.

──────────────────────────────

1️⃣ TRANSFER / ACCESS LOG

├── Definition
│   • Records every request
│     made to the server.
│
├── Stores
│   • Page visits
│   • File downloads
│   • Image requests
│   • URL access
│
├── Purpose
│   • Track website traffic.
│
└── Example
    User opens:
    www.website.com/home

    Entry recorded in Access Log.

──────────────────────────────

2️⃣ AGENT LOG

├── Definition
│   • Records User Agent information.
│
├── User Agent Means
│   • Browser being used.
│   • Operating System being used.
│
├── Examples
│
│   Browser:
│     Chrome
│     Firefox
│     Safari
│     Edge
│
│   Operating System:
│     Windows
│     Linux
│     Android
│     iOS
│
└── Purpose
    • Understand visitor technology.

──────────────────────────────

3️⃣ REFERRER LOG

├── Definition
│   • Records previous webpage
│     visited by user.
│
├── Answers Question
│
│   "Where did user come from?"
│
├── Example
│
│   User clicks:
│   Google Search
│       ↓
│   Your Website
│
│   Referrer:
│   google.com
│
└── Purpose
    • Traffic source analysis.
    • Marketing analysis.

──────────────────────────────

4️⃣ ERROR LOG

├── Definition
│   • Records server and website errors.
│
├── Examples
│
│   404 Error
│   Page Not Found
│
│   500 Error
│   Internal Server Error
│
│   Database Failure
│
│   Server Crash
│
└── Purpose
    • Troubleshooting.
    • Website maintenance.
    • Error monitoring.

──────────────────────────────

📈 5. WHY WEB USAGE LOGS ARE IMPORTANT?
──────────────────────────────

├── Understand User Behavior
│
├── Measure Website Traffic
│
├── Improve Website Design
│
├── Detect Errors
│
├── Support Marketing Decisions
│
├── Customer Analysis
│
└── Personalized Recommendations

──────────────────────────────

📝 SHORT NOTES (EXAM READY)
──────────────────────────────

Access Log
• Stores all requests made to server.
• Tracks visits, downloads and page access.

Agent Log
• Stores browser and operating system details.
• Helps identify visitor technology.

Referrer Log
• Stores source website from where user arrived.
• Useful for traffic analysis.

Error Log
• Stores website and server errors.
• Helps detect and fix problems.

──────────────────────────────

🎯 MOST IMPORTANT EXAM QUESTIONS

1. Define Web Mining.

2. Differentiate between
   Traditional Data Mining
   and Web Mining.

3. Explain three classes
   of Web Mining.

4. What is Web Usage Mining?

5. What are Web Logs?

6. Explain Access Log.

7. Explain Agent Log.

8. Explain Referrer Log.

9. Explain Error Log.

10. What are data sources
    for Web Usage Mining?

──────────────────────────────

🧠 LAST-MINUTE MEMORY TRICK

WEB MINING
│
├── Structure Mining
│   └── Hyperlinks
│
├── Content Mining
│   ├── Classification
│   ├── Clustering
│   └── Association
│
└── Usage Mining
    └── LOGS
         │
         ├── Access Log
         ├── Agent Log
         ├── Referrer Log
         └── Error Log
```

Outlier & Anomaly Detection.txt

```
🚨 OUTLIER (ANOMALY) DETECTION
│
├──────────────────────────────────────────
│ 1. INTRODUCTION
├──────────────────────────────────────────
│
│
├── Definition
│   • An outlier is a data object that
│     deviates significantly from the
│     majority of the dataset.
│
│   • Appears abnormal, unusual,
│     rare or suspicious.
│
├── Other Names
│   • Anomaly
│   • Exception
│   • Deviation
│
├── Goal of Outlier Detection
│   • Identify abnormal patterns.
│   • Detect unusual events.
│   • Discover rare but important cases.
│
└── Real-Life Examples
    • Credit Card Fraud
    • Network Intrusion Detection
    • Medical Diagnosis
    • Insurance Fraud
    • Equipment Failure Detection

──────────────────────────────────────────

⚠ 2. OUTLIERS VS NOISE
──────────────────────────────────────────

├── Noise
│
│   • Random error.
│   • Small variation.
│   • Usually ignored or removed.
│
│   Example:
│     Customer buys one extra coffee.
│
└── Outlier
    • Important abnormal event.
    • Requires investigation.
    • May indicate fraud or attack.

    Example:
      Customer suddenly spends
      $50,000 in another country.

⭐ Exam Point:
Noise = Error
Outlier = Interesting Information

──────────────────────────────────────────

📌 3. TYPES OF OUTLIERS
   (VERY IMPORTANT)
──────────────────────────────────────────

├── 1. Global Outlier
│   (Point Anomaly)
│
│   Definition:
│   • A single point that is far away
│     from all other data points.
│
│   Characteristics:
│   • Clearly abnormal.
│   • Easy to identify.
│
│   Example:
│
│   Heights:
│   5.2, 5.4, 5.5, 5.7, 5.8, 7.5
│
│   → 7.5 is Global Outlier
│
├── 2. Contextual Outlier
│   (Conditional Outlier)
│
│   Definition:
│   • Outlier only in a specific context.
│
│   Context Examples:
│     Time
│     Location
│     Season
│
│   Example:
│
│   Temperature = 28°C
│
│   Summer → Normal
│   Winter → Outlier
│
│   Same value,
│   different context.
│
├── 3. Collective Outlier
│
│   Definition:
│   • Individual objects may be normal.
│   • Entire group becomes abnormal.
│
│   Example:
│
│   One delayed order → Normal
│
│   100 delayed orders together
│   → Collective Outlier
│
│   Indicates:
│     Supply chain failure
│     System malfunction
│
└── Memory Trick:
    Global = One abnormal point
    Contextual = Abnormal in context
    Collective = Abnormal group

──────────────────────────────────────────

⚡ 4. APPLICATIONS OF OUTLIER DETECTION
──────────────────────────────────────────

├── Fraud Detection
│   • Credit card fraud
│   • Insurance fraud
│
├── Cybersecurity
│   • Intrusion detection
│   • Network attack detection
│
├── Medical Analysis
│   • Disease detection
│   • Abnormal patient behavior
│
├── Manufacturing
│   • Defective products
│   • Machine failures
│
└── Financial Analysis
    • Suspicious transactions
    • Market manipulation detection

──────────────────────────────────────────

❓ 5. CHALLENGES IN OUTLIER DETECTION
──────────────────────────────────────────

├── Blurry Boundary Problem
│
│   • Hard to define exact line
│     between normal and abnormal.
│
├── Application Dependency
│
│   • Outlier definition changes
│     from one domain to another.
│
│   Example:
│     Medical systems need strict
│     anomaly detection.
│
├── Noise Handling
│
│   • Noise may look like outliers.
│   • Outliers may hide inside noise.
│
├── Data Quality Issues
│
│   • Missing values.
│   • Incorrect data.
│
└── Explainability
    • System should explain WHY
      object was marked as outlier.

──────────────────────────────────────────

🧠 6. LEARNING APPROACHES
──────────────────────────────────────────

├── 1. Supervised Methods
│
│   Definition:
│   • Training data already labeled.
│
│   Labels:
│     Normal
│     Outlier
│
│   Advantages:
│   • High accuracy.
│
│   Limitation:
│   • Hard to obtain enough
│     outlier examples.
│
├── 2. Unsupervised Methods
│
│   Definition:
│   • No labels available.
│
│   Assumption:
│   • Normal data forms clusters.
│   • Isolated points are outliers.
│
│   Advantages:
│   • Most widely used.
│
│   Limitation:
│   • May produce false alarms.
│
├── 3. Semi-Supervised Methods
│
│   Definition:
│   • Few labeled examples available.
│
│   Uses:
│     Small labeled data
│     + Large unlabeled data
│
│   Goal:
│     Learn normal behavior.
│
└── Memory Trick:
    Supervised = Fully labeled
    Semi-supervised = Partially labeled
    Unsupervised = No labels

──────────────────────────────────────────

📊 7. OUTLIER DETECTION TECHNIQUES
──────────────────────────────────────────

                OUTLIER DETECTION
                        │
         ┌──────────────┼──────────────┐
         ▼              ▼              ▼
Statistical        Proximity-Based      Clustering-Based
(Model-Based)

──────────────────────────────────────────

📈 A. STATISTICAL APPROACHES
   (MODEL-BASED)
──────────────────────────────────────────

├── Idea
│   • Assume normal data follows
│     a statistical distribution.
│
│   • Objects violating the model
│     become outliers.
│
├── Parametric Methods
│
│   Definition:
│   • Assume specific distribution.
│
│   Example:
│     Normal Distribution
│     Bell Curve
│
│   Important Techniques:
│
│   ├── Boxplot
│   │   • Values outside whiskers
│   │     are outliers.
│   │
│   └── 3-Sigma Rule
│       • Mean ± 3 Standard Deviations
│       • Outside this range = Outlier
│
└── Non-Parametric Methods
    │
    Definition:
    • No predefined distribution.

    Example:
      Histogram

    Rule:
      Rare or empty bins indicate
      potential outliers.

──────────────────────────────────────────

📍 B. PROXIMITY-BASED APPROACHES
──────────────────────────────────────────

├── Idea
│   • Examine nearby neighbors.
│
├── Distance-Based Methods
│
│   Definition:
│   • Count neighbors within radius.
│
│   Rule:
│   • Few neighbors → Outlier
│
│   Characteristics:
│   • Isolated objects are suspicious.
│
└── Density-Based Methods
    │
    Definition:
    • Compare local density.

    Rule:
    • Lower density than neighbors
      → Outlier

    Example:
      Object in empty region while
      neighbors are densely packed.

──────────────────────────────────────────

🎯 C. CLUSTERING-BASED APPROACHES
──────────────────────────────────────────

├── Idea
│   • Group similar objects
│     into clusters.
│
├── Assumption
│   • Normal data belongs to
│     large dense clusters.
│
├── Outliers Are
│
│   • Objects in very small clusters
│
│   OR
│
│   • Objects not belonging to
│     any cluster
│
└── Example
    Cluster A = 1000 points
    Cluster B = 2 points

    → Cluster B likely contains
      outliers

──────────────────────────────────────────

⭐ COMPARISON OF TECHNIQUES
──────────────────────────────────────────

Statistical Approach
• Uses probability distributions.
• Example: 3-Sigma Rule, Boxplot.

Distance-Based Approach
• Uses neighbor distance.
• Isolated points are outliers.

Density-Based Approach
• Uses local density.
• Sparse region = outlier.

Clustering-Based Approach
• Uses clusters.
• Small clusters = outliers.

──────────────────────────────────────────

🎯 MOST IMPORTANT EXAM QUESTIONS
──────────────────────────────────────────

1. Define Outlier.
2. Difference between Outlier and Noise.
3. Explain Global Outlier.
4. Explain Contextual Outlier.
5. Explain Collective Outlier.
6. Applications of Outlier Detection.
7. Challenges in Outlier Detection.
8. Supervised vs Unsupervised Methods.
9. Statistical Outlier Detection.
10. Parametric vs Non-Parametric Methods.
11. Explain 3-Sigma Rule.
12. Explain Boxplot.
13. Distance-Based Outlier Detection.
14. Density-Based Outlier Detection.
15. Clustering-Based Outlier Detection.
```

Heirarchical.txt

```
📊 CLUSTERING METHODS

────────────────────────────────────────
1. HIERARCHICAL CLUSTERING
────────────────────────────────────────

├── Definition
│   • Creates a hierarchy (tree)
│     of clusters.
│
│   • Result represented by:
│     DENDROGRAM (Tree Structure)
│
├── Advantages
│   • Easy visualization.
│   • Good for summarization.
│   • Reveals hidden structures.
│   • Shows relationships among clusters.
│
├── Types
│   │
│   ├── Agglomerative
│   │   (Bottom-Up)
│   │
│   │   Process:
│   │   Step 1 → Each object =
│   │            separate cluster
│   │
│   │   Step 2 → Merge closest clusters
│   │
│   │   Step 3 → Continue merging
│   │
│   │   Step 4 → One root cluster remains
│   │
│   │   Example:
│   │   AGNES
│   │   (Agglomerative Nesting)
│   │
│   └── Divisive
│       (Top-Down)
│
│       Process:
│       Step 1 → One big cluster
│
│       Step 2 → Split into subclusters
│
│       Step 3 → Continue splitting
│
│       Step 4 → Small clusters formed
│
│       Example:
│       DIANA
│       (Divisive Analysis)
│
└── Memory Trick
    AGNES = Merge
    DIANA = Divide

────────────────────────────────────────

✏️ 2. LINKAGE METHODS
   (Distance Between Clusters)
────────────────────────────────────────

├── Single Linkage
│   (Minimum Distance)
│
│   Definition:
│   • Distance between closest
│     pair of objects.
│
│   Characteristics:
│   • Forms chain-like clusters.
│   • Sensitive to noise.
│
│   Example:
│   Minimum Spanning Tree concept.
│
├── Complete Linkage
│   (Maximum Distance)
│
│   Definition:
│   • Distance between farthest
│     pair of objects.
│
│   Characteristics:
│   • Compact clusters.
│   • Small cluster diameter.
│
├── Average Linkage
│
│   Definition:
│   • Average distance between
│     all objects.
│
│   Characteristics:
│   • Balanced approach.
│
└── Mean Linkage
    • Uses mean distances.
    • Less sensitive to outliers.
────────────────────────────────────────

⭐ IMPORTANT COMPARISON

Single Linkage
• Closest points
• Chain effect
• Sensitive to noise

Complete Linkage
• Farthest points
• Compact clusters

Average Linkage
• Average distance
• Balanced performance

────────────────────────────────────────

🚀 3. ADVANCED HIERARCHICAL METHODS
────────────────────────────────────────

Basic Hierarchical Problems:

• Cannot undo merge decisions
• Cannot undo split decisions
• Poor scalability for huge datasets

Solution:
Advanced Multiphase Methods

────────────────────────────────────────

🌲 A. BIRCH
(Balanced Iterative Reducing and
 Clustering using Hierarchies)
────────────────────────────────────────

├── Purpose
│   • Designed for very large datasets.
│   • Mainly numeric data.
│
├── Main Idea
│   • Summarize clusters using
│     Clustering Features (CF).
│
├── CF Vector
│
│   CF = < n , LS , SS >
│
│   n  = Number of points
│   LS = Linear Sum
│   SS = Square Sum
│
├── Data Structure
│
│   CF Tree
│
│   Purpose:
│   Compact representation
│   of huge datasets.
│
├── Phases
│
│   Phase 1:
│   • Scan data.
│   • Build CF Tree.
│
│   Phase 2:
│   • Apply clustering algorithm
│     on leaf nodes.
│
├── Time Complexity
│
│   O(n)
│
├── Advantages
│
│   • Very fast.
│   • Scalable.
│   • Memory efficient.
│
└── Limitation
    • Poor for non-spherical clusters.
────────────────────────────────────────

🦎 B. CHAMELEON
────────────────────────────────────────

├── Main Idea
│
│   • Uses k-Nearest Neighbor Graph.
│
├── Process
│
│   Step 1:
│   Create sparse graph.
│
│   Step 2:
│   Create subclusters.
│
│   Step 3:
│   Merge similar subclusters.
│
├── Uses Two Measures
│
│   • Relative Interconnectivity
│
│   • Relative Closeness
│
└── Advantage
    • Finds complex cluster shapes.
────────────────────────────────────────

🎲 C. PROBABILISTIC
   HIERARCHICAL CLUSTERING
────────────────────────────────────────

├── Main Idea
│
│   • Uses probability models.
│
│   • Uses generative models.
│
├── Example
│
│   Gaussian Distribution
│
├── Goal
│
│   Maximize likelihood
│   of data fitting model.
│
├── Advantages
│
│   • Handles uncertainty.
│   • Handles missing values.
│
└── Application
    • Incomplete datasets.

────────────────────────────────────────

🌊 4. DENSITY-BASED CLUSTERING
────────────────────────────────────────

├── Definition
│
│   • Clusters are dense regions.
│
│   • Sparse regions separate clusters.
│
├── Major Advantage
│
│   • Finds arbitrary-shaped clusters.
│
│   • Handles noise well.
│
└── Main Algorithms
    • DBSCAN
    • OPTICS
    • DENCLUE

────────────────────────────────────────

🔥 A. DBSCAN
(Density-Based Spatial Clustering
 of Applications with Noise)
────────────────────────────────────────

├── Parameters
│
│   ε (Epsilon)
│   • Radius
│
│   MinPts
│   • Minimum neighbors required
│
├── Core Object
│
│   Definition:
│   Object having at least
│   MinPts neighbors within ε.
│
├── Concepts
│
│   • Core Point
│   • Border Point
│   • Noise Point
│
├── Connectivity
│
│   • Density Reachable
│
│   • Density Connected
│
├── Advantages
│
│   • Finds arbitrary shapes.
│
│   • Handles noise.
│
├── Time Complexity
│
│   With Index:
│   O(n log n)
│
│   Without Index:
│   O(n²)
│
└── Limitation
    • Sensitive to parameter selection.
────────────────────────────────────────

✅ B. OPTICS
(Ordering Points To Identify
 Clustering Structure)
────────────────────────────────────────

├── Purpose
│
│   • Improvement over DBSCAN.
│
├── Problem Solved
│
│   • DBSCAN sensitive to
│     global parameters.
│
├── Output
│
│   • Cluster Ordering
│
│   Instead of strict clusters.
│
├── Important Terms
│
│   Core Distance
│   • Distance needed for point
│     to become core object.
│
│   Reachability Distance
│   • Distance needed to make
│     point reachable.
│
└── Advantage
    • Detects clusters
      with varying densities.
────────────────────────────────────────

⛰️ C. DENCLUE
(DENsity-based CLUstEring)
────────────────────────────────────────

├── Main Idea
│
│   • Uses mathematical
│     density functions.
│
├── Method
│
│   Kernel Density Estimation
│
│   Example:
│   Gaussian Kernel
│
├── Cluster Centers
│
│   Density Attractors
│
│   (Local Maximum Density Points)
│
├── Process
│
│   Hill Climbing Procedure
│
│   → Move toward highest density.
│
└── Advantage
    • Strong mathematical foundation.
────────────────────────────────────────

🟪 5. GRID-BASED CLUSTERING
────────────────────────────────────────

├── Definition
│
│   • Divides data space
│     into grid cells.
│
├── Main Idea
│
│   Space Driven
│
│   Instead of:
│   Data Driven
│
├── Advantage
│
│   • Very fast processing.
│
│   • Works well with huge datasets.
│
└── Main Algorithms
    • STING
    • CLIQUE

────────────────────────────────────────

🏢 A. STING
(Statistical Information Grid)
────────────────────────────────────────

├── Main Idea
│
│   • Divide space into
│     hierarchical rectangular cells.
│
├── Information Stored
│
│   Each cell stores:
│
│   • Count
│   • Mean
│   • Minimum
│   • Maximum
│   • Standard Deviation
│
├── Advantages
│
│   • Fast query processing.
│
│   • Efficient clustering.
│
├── Complexity
│
│   Clustering:
│   O(n)
│
│   Querying:
│   O(g)
│
│   g = Number of grid cells
│
└── Suitable For
    • Spatial databases.

────────────────────────────────────────

🧩 B. CLIQUE
────────────────────────────────────────

├── Full Form
│
│   Clustering In QUEst
│
├── Type
│
│   • Grid-Based
│   • Density-Based
│
├── Purpose
│
│   • High-dimensional data.
│
├── Main Idea
│
│   Dense high-dimensional cell
│   exists only if all lower-
│   dimensional projections
│   are dense.
│
├── Uses
│
│   Apriori Principle
│
├── Additional Principle
│
│   MDL
│   (Minimum Description Length)
│
└── Advantage
    • Discovers clusters in
      high-dimensional subspaces.
────────────────────────────────────────

⭐ ALGORITHM COMPARISON
────────────────────────────────────────

AGNES
• Hierarchical
• Bottom-Up
• Merge clusters

DIANA
• Hierarchical
• Top-Down
• Split clusters

BIRCH
• Hierarchical
• Uses CF Tree
• O(n)

CHAMELEON
• k-NN Graph
• Relative Interconnectivity

DBSCAN
• Density-Based
• Uses ε and MinPts

OPTICS
• Improved DBSCAN
• Uses Core Distance

DENCLUE
• Kernel Density
• Density Attractors

STING
• Grid-Based
• Hierarchical Cells

CLIQUE
• Grid + Density
• High-Dimensional Data
```

Supervised and Unsupervised Learning.txt

```
🤖 2. MACHINE LEARNING OVERVIEW
────────────────────────────────

├── Definition
│
│   • Field of Artificial Intelligence
│     that enables computers to learn
│     from data without being explicitly
│     programmed.
│
├── Main Categories
│
│       Machine Learning
│              │
│      ┌───────┴────────┐
│      ▼                ▼
│
│  Supervised      Unsupervised
│   Learning         Learning
│
└── Purpose
    • Learn patterns from data.
    • Make predictions.
    • Discover hidden relationships.

────────────────────────────────
🎯 3. SUPERVISED LEARNING
────────────────────────────────

├── Definition
│
│   • Learning from labeled data.
│
│   • Training data already contains
│     correct answers (labels).
│
│   • Acts like a teacher guiding
│     a student.
│
├── Main Idea
│
│   Input Data + Correct Answers
│             ↓
│         Training
│             ↓
│           Model
│             ↓
│        Prediction
│
├── Workflow
│
│   Step 1:
│   Collect labeled data
│
│   Step 2:
│   Train model
│
│   Step 3:
│   Test model
│
│   Step 4:
│   Generate predictions
│
└── Goal
    • Predict unknown outputs.

────────────────────────────────
📈 4. TYPES OF SUPERVISED LEARNING
────────────────────────────────

├── A. REGRESSION
│
│   Definition:
│   • Predicts continuous numeric values.
│
│   Target Variable:
│   • Real-valued (continuous)
│
│   Examples:
│   • House Price Prediction
│   • Temperature Prediction
│   • Stock Price Prediction
│   • Sea Ice Extent Prediction
│
│   Output Examples:
│   • 45.6
│   • 100.2
│   • 250000
│
└── B. CLASSIFICATION

    Definition:
    • Predicts categories/classes.

    Target Variable:
    • Categorical

    Examples:
    • Spam / Not Spam
    • Pass / Fail
    • Disease / Healthy
    • Benign / Malignant Tumor

    Output Examples:
    • Yes / No
    • 0 / 1
    • Class A / B / C

────────────────────────────────
⭐ REGRESSION VS CLASSIFICATION

Regression
• Numeric Output
• Continuous Values

Examples:
House Price
Temperature

Classification
• Categorical Output
• Class Labels

Examples:
Spam Detection
Cancer Classification

────────────────────────────────
⚙️ 5. SUPERVISED LEARNING ALGORITHMS
────────────────────────────────

├── Linear Regression
│
│   • Predicts continuous values.
│
│   Example:
│   House price prediction.
│
├── Logistic Regression
│
│   • Classification algorithm.
│
│   Example:
│   Disease prediction.
│
├── Decision Trees
│
│   • Tree-like decision structure.
│
│   Example:
│   Loan approval.
│
├── Support Vector Machine (SVM)
│
│   • Separates classes using
│     optimal boundaries.
│
│   Example:
│   Image classification.
│
└── Neural Networks
    • Brain-inspired model.
    • Used in Deep Learning.
    • Handles complex patterns.

────────────────────────────────
✅ 6. ADVANTAGES OF SUPERVISED LEARNING
────────────────────────────────

├── High Accuracy
│
├── Easy Evaluation
│
├── Clear Learning Process
│
├── Strong Prediction Capability
│
└── Works Well When Labels Exist

────────────────────────────────
X 7. DISADVANTAGES OF SUPERVISED LEARNING
────────────────────────────────

├── Requires Labeled Data
│
├── Labeling Is Expensive
│
├── Data Preparation Takes Time
│
├── Large Training Data Needed
│
└── Human Effort Required

────────────────────────────────
🔍 8. UNSUPERVISED LEARNING
────────────────────────────────

├── Definition
│
│   • Learning from unlabeled data.
│
│   • No teacher or supervision.
│
│   • System discovers patterns itself.
│
├── Main Idea
│
│   Raw Data
│      ↓
│   Discover Structure
│      ↓
│   Hidden Patterns Found
│
├── Goal
│
│   • Find hidden relationships.
│
│   • Group similar objects.
│
│   • Compress data.
│
│   • Discover structures.
│
└── Characteristics
    • No labels.
    • Exploratory analysis.
    • Pattern discovery.

────────────────────────────────
🎯 9. TYPES OF UNSUPERVISED LEARNING
────────────────────────────────

├── Clustering
│
│   • Group similar objects.
│
├── Association
│
│   • Find relationships
│     between items.
│
│   Example:
│   Bread → Butter
│
├── Dimensionality Reduction
│
│   • Reduce number of features.
│
│   Example:
│   PCA

────────────────────────────────
✅ 10. ADVANTAGES OF UNSUPERVISED LEARNING
────────────────────────────────

├── No Labeling Needed
│
├── Saves Time
│
├── Discovers Hidden Patterns
│
├── Useful for Exploration
│
└── Handles Large Data

────────────────────────────────
X 11. DISADVANTAGES OF UNSUPERVISED LEARNING
────────────────────────────────

├── Lower Accuracy
│
├── Hard to Evaluate
│
├── Results May Be Ambiguous
│
├── No Ground Truth Available
│
└── Interpretation Can Be Difficult

────────────────────────────────
📊 12. CLUSTERING (DEEP DIVE)
────────────────────────────────

├── Definition
│
│   • Most popular unsupervised
│     learning technique.
│
│   • Groups similar objects together.
│
│   • Separates dissimilar objects.
│
├── Objective
│
│   Same Cluster
│   → High Similarity
│
│   Different Clusters
│   → Low Similarity
│
├── Nature
│
│   • Exploratory Data Analysis.
│
│   • Pattern Discovery.
│
│   • No labels required.
│
└── Goal
    • Reveal hidden structures.

────────────────────────────────
🌍 13. APPLICATIONS OF CLUSTERING
────────────────────────────────

├── Biology
│
│   • Species classification.
│
│   • Genetic analysis.
│
├── Medical Imaging
│
│   • Tissue identification.
│
│   • Disease detection.
│
├── Market Research
│
│   • Customer segmentation.
│
│   • Consumer behavior analysis.
│
├── Recommender Systems
│
│   • Amazon recommendations.
│
│   • Netflix movie suggestions.
│
└── Business Intelligence
    • Customer profiling.
    • Marketing campaigns.

────────────────────────────────
⚙️ 14. TYPES OF CLUSTERING ALGORITHMS
────────────────────────────────

                   CLUSTERING
                       │
      ┌────────────────┼────────────────┐
      ▼                ▼                ▼

Partition-Based   Hierarchical      Density-Based

────────────────────────────────
📌 A. PARTITION-BASED CLUSTERING

├── Definition
│   • Divides data into k clusters.
│
├── Characteristics
│   • Fast
│   • Efficient
│
└── Example
    • K-Means

────────────────────────────────
🌳 B. HIERARCHICAL CLUSTERING

├── Definition
│   • Produces tree-like clusters.
│
├── Output
│   • Dendrogram
│
├── Types
│
│   • Agglomerative
│     (Bottom-Up)
│
│   • Divisive
│     (Top-Down)
│
└── Advantage
    • Easy visualization.

────────────────────────────────
🌊 C. DENSITY-BASED CLUSTERING

├── Definition
│   • Forms clusters from dense regions.
│
├── Characteristics
│   • Finds arbitrary shapes.
│
│   • Handles noise well.
│
└── Example
    • DBSCAN

────────────────────────────────
⭐ SUPERVISED VS UNSUPERVISED
────────────────────────────────

Supervised Learning
│
├── Labeled Data
├── Teacher Available
├── Predict Output
├── Higher Accuracy
└── Examples:
    Regression
    Classification

Unsupervised Learning
│
├── Unlabeled Data
├── No Teacher
├── Discover Patterns
├── Exploratory Analysis
└── Examples:
    Clustering
    Association
    Dimensionality Reduction
```